

RPS-1 Autonomous Hand Mechanical Design

MSE 3302B - Sensors and Actuators

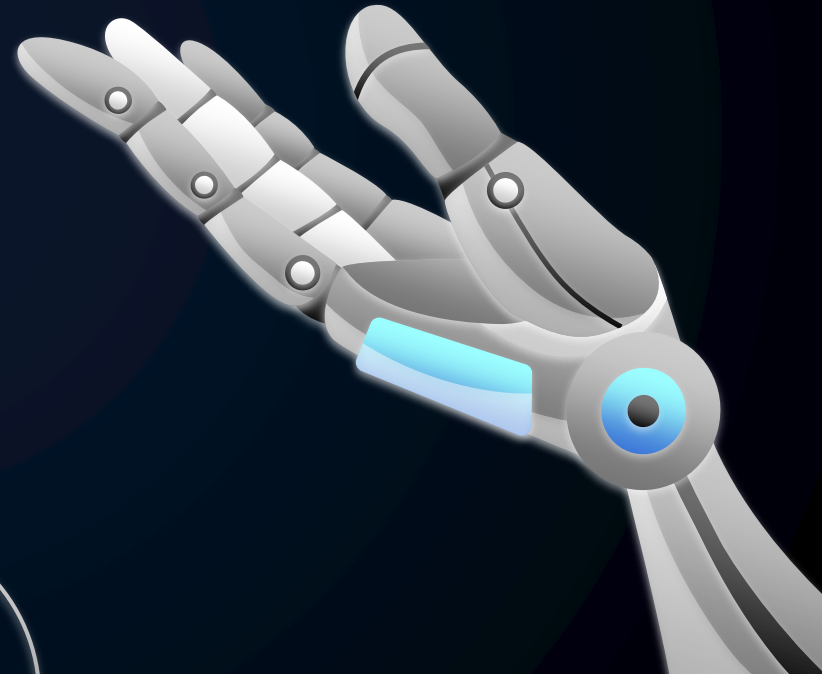


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Design Objectives & Engineering Constraints



PRIMARY OBJECTIVES

Produce stable Rock, Paper, and Scissors gesture outputs

Ensure repeatable transitions between gestures

Simplify manufacturing, assembly, and maintenance



ENGINEERING CONSTRAINTS

Limited actuator count and wiring complexity

Need for quick iteration and debugging during prototyping

Mechanical robustness for repeated demonstrations

Mechanical Development Process



Phase 1: CAD Concept

Built the initial palm and finger architecture

Verified basic range of motion and finger articulation



Phase 2: Mechanical Iteration

Adjusted joint clearances and contact geometry

Tuned stop angles for consistent final finger positions

Improved part fit for smoother assembly



Phase 3: Integration

Added servo mounts and routing considerations

Updated geometry for easier part replacement and maintenance

Design Decisions

Hand Configuration

Criterion	Weight	String-Pulley System	Direct Drive	4-Bar Linkage
Actuation Speed	0.35	0	1	1
Precision	0.4	1	0	-1
Manufacturing Complexity	0.2	1	-1	0
Cost	0.05	1	1	1
Total	1	0.65	0.2	0

Finger Retention

Criterion	Weight	Elastic Bands	Compression Springs	Opposing Motor
Motor Count	0.4	1	1	-1
Manufacturability	0.35	1	-1	-1
Reliability	0.25	0	1	1
Total	1	0.75	0.3	-0.5

Actuation Architecture

Why grouped actuation:

- Lower actuator count
- Reduced wiring and power requirements
- Better reliability for this specific task



Set 1
Pinky + Ring +
Thumb

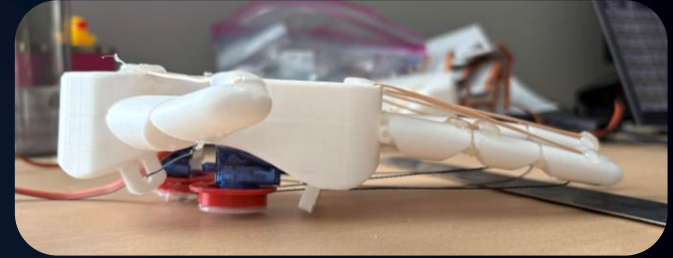
Set 2
Index + Middle

Electronics Selection



- Built-in position control
- Suitable torque for finger motion
- Simple command interface
- Fast integration and testing

FS90R SERVO MOTOR

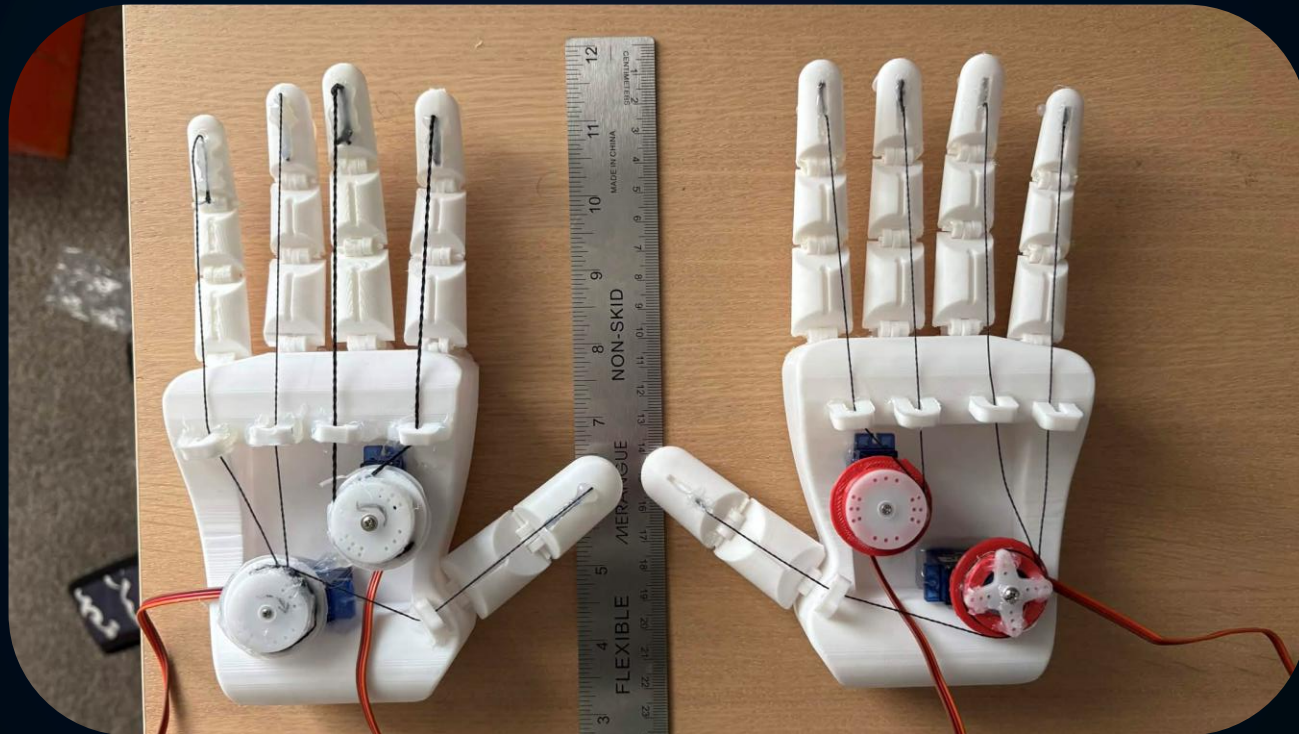


- Reliable PWM generation for servo control
- Easy debugging workflow
- Familiar and stable platform
- Sufficient I/O for grouped actuation design

Arduino Uno R3



Final Mechanical Outcome



Final Mechanical Outcome

